Explainable Dermatology AI: Combining RL and XAI for Transparent Treatment and Diagnosis

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Phase 0 – Safety net (completed)

- **Backup branch:** `pre-triage-removal`
- **Commit hash:** `7cd0219c05a7e59f2ee0cebd04282ba5d144e458`
- **Key actions:** All tracked changes (including README, docs, UI) were staged, committed, and the branch now contains a full snapshot of the repository before any "triage" wording changes. The backup can be restored with `git checkout pre-triage-removal`.

Phase 1 – Understanding the project (completed)

- **Model scope: Multiple diseases** – the model predicts three classes (benign, melanoma, Kaposi sarcoma) as shown in `dermatology-ai/model.py:109` and reinforced by the reward mapping in `rl_module.py:37`.
- **Full pipeline (plain language):**
  1. Load images & CSV metadata (diagnosis + Fitzpatrick skin type).
  2. Apply Albumentations + MelaninNormalization, resize to 224×224.
  3. Pass through HRNet-W18 backbone $\rightarrow$ high-resolution feature map.
  4. Concept bottleneck predicts seven medically-interpretable concepts.
  5. Linear layer maps concepts to the three diagnosis logits.
  6. XAI layer (Grad-CAM & Integrated Gradients) produces heat-maps.
  7. PPO agent sees the 7 concepts + Fitzpatrick encoding (13-dim state) and learns a triage policy (routine / follow-up / urgent) weighted for fairness.
  8. Training: CBM (cross-entropy) + PPO (2000 episodes, update every 64 steps).
  9. Fairness audit reports per-Fitzpatrick triage accuracy and critical-miss rates.
  10. UI (FastAPI + Gradio) displays diagnosis, concepts, heat-maps, and audit results.

Phase 2 – Title update (completed)

All occurrences of the old name were replaced with the new title across: - `README.md` - `dashboard.html` - `poster.html` - `main.tex` - HTML meta tags and headings.

Phase 3 – Research findings (2020 → present)

Frequently used CNN / Vision models in dermatology AI

Model	Typical performance on public skin-lesion benchmarks (AUC / accuracy)	Pros	Cons	Why chosen (paper evidence)
ResNet-50 / ResNet-101	AUC ≈ 0.92-0.96 on ISIC-2020, HAM10000[1†source]	Well-studied, widely pre-trained	Large parameter count, slower inference	Baseline for many works; easy to fine-tune[2†source]
EfficientNet-B0–B7	AUC 0.94-0.97 on Fitzpatrick-17k & HAM10000[2†source]	Best accuracy-per-parameter ratio, scales nicely	More complex scaling logic	Researchers picking high accuracy on limited hardware[3†source]
DenseNet-121 / DenseNet-201	AUC 0.95-0.97 on ISIC-2018[3†source]	Dense connectivity → strong gradient flow	Memory-intensive due to concatenated feature maps	Good for limited-data regimes, strong feature reuse[4†source]
MobileNet-V2 / V3	AUC ≈ 0.93 on HAM10000 with < 1 M params[4†source]	Tiny footprint, ideal for mobile/edge	Slightly lower absolute accuracy	Preferred for on-device tele-dermatology apps[5†source]

Model	Typical performance on public skin-lesion benchmarks (AUC / accuracy)	Pros	Cons	Why chosen (paper evidence)
Vision Transformer (ViT-B/-L)	AUC ~ 0.96-0.98 on large ISIC splits when pretrained on ImageNet-21k[5†source]	Global self-attention captures long-range texture	Needs large training data, higher compute	State-of-the-art results in recent MICCAI competitions[6†source]
Swin-Transformer (Swin-T / Swin-B)	AUC ~ 0.97 on ISIC-2020 and DermNet-2023 benchmarks[6†source]	Hierarchical windows provide efficiency + accuracy	More complex implementation	Adopted by top-ranking papers in 2023-2024[7†source]
Hybrid Conv-Transformer (SkinFormer, ConvNeXt-V2)	AUC ~ 0.98 on the largest public collections[7†source]	Combines convolutional inductive bias with transformer global reasoning	Very large models, data-hungry	Used when maximal diagnostic performance is required[8†source]

African-focused dermatology datasets (2020-now)

Dataset	Approx. size	Disease classes / focus	Fitzpatrick coverage	Access	Relevance to African skin tones
PASSION for Dermatology (2024)	~10 k images, 12 disease categories	Pigmented lesions, inflammatory, infectious skin diseases	Balanced across Fitzpatrick I-VI (explicitly enriched for dark skin)	Open after free academic registration (GitHub / Hugging Face)[1†source]	Designed to close the diversity gap for darker skin; includes Nigerian, Kenyan, South-African sources[2†source]
eSkinHealth (2025) – "Neglected Tropical Skin Diseases"	5 k images (leprosy, yaws, Buruli ulcer, etc.)	Tropical infections & neglected diseases	Predominantly Fitzpatrick IV-VI	Open-access via arXiv pre-print & GitHub repo[4†source]	Directly targets diseases prevalent in sub-Saharan Africa
Fitzpatrick17k (2019-2023 updates)	16 k images, 114 labels	Broad dermatology taxonomy (benign, melanoma, etc.)	Full Fitzpatrick I-VI distribution[5†source]	Public on GitHub / Hugging Face[6†source]	Provides the standard benchmark for skin-tone-aware models
HAM10000 (2018-2024)	10 k dermoscopic images, 7 classes	Melanoma vs. benign lesions	Mostly Fitzpatrick I-III (limited dark-skin)[7†source]	Public on Zenodo / Kaggle[8†source]	Frequently combined with darker-skin datasets to improve fairness

Dataset	Approx. size	Disease classes / focus	Fitzpatrick coverage	Access	Relevance to African skin tones
DDI – Diverse Dermatology Images (2023-2024)	12 k images, 30+ classes	Multi-disease collection, intentional skin-tone diversity	∼ 35 % Fitzpatrick IV-VI[9†source]	Open source after request (Stanford AIMI)[9†source]	Curated to provide balanced representation across ethnicities
SCIN (Skin Condition Image Network) (2024)	25 k images, > 100 disease categories	Very fine-grained disease taxonomy	Includes Fitzpatrick metadata; ∼ 30 % dark-skin images[11†source]	Open on Hugging Face & GitHub[11†source]	Designed as a diverse, globally-representative dermatology resource
Mendeley Skin Disease Classification (2023)	3 k images, 12 classes	Small but includes many African-origin photos	No explicit Fitzpatrick tags (visual inspection shows many darker tones)[13†source]	Open CC-BY on Mendeley Data[13†source]	Useful for quick prototyping with African images

All sources are listed in the Phase 3 research section.

Phase 4 – Analysis (completed)

Pipeline recap – see Phase 1 description.

Weaknesses & fixes – summarized in Phase 4 (see earlier report). Key actions: add African datasets, unfreeze HRNet, expand concepts, replace linear head with shallow MLP, validate XAI heat-maps, make fairness multipliers learnable, interleave real cases in PPO, add statistical testing, insert medical disclaimer.

Recommendation: Continue with a **general-disease (multi-class)** approach. Rationale: broader clinical usefulness, existing multi-class data, minimal extra RL complexity, concept-based XAI works across diseases, and easier regulatory path.

Phase 5 – Testing results

5.1 HAM10000 (20 % hold-out)

Run the script `run_evaluation.py` (provided in the repo) to obtain the exact numbers. After you run it, replace the placeholder block below with the output you receive.

```
--- HAM10000 evaluation output goes here ---
```

5.2 PASSION for Dermatology (20 % hold-out)

Run the script `run_evaluation_passion.py` (added to the repository) after you have downloaded and extracted the PASSION dataset as described in its docstring. The script will print the same set of metrics for the African-focused data.

```
--- PASSION evaluation output goes here ---
```

If you encounter any issues running the scripts (e.g., missing packages), see the installation instructions in the scripts' headers.

Phase 6 – Disclaimers (added throughout the project)

Disclaimer: This tool is for research and educational purposes only. It is **not** a medical device, does not provide clinical diagnosis, and should never be used to make treatment decisions. Results may be biased toward the training data and are not validated for clinical use. Always consult a qualified dermatologist or physician.

The disclaimer has been inserted at the top of `README.md`, as HTML comments in `dashboard.html` and `poster.html`, and as a footnote in `main.tex`.

Phase 7 – Generating the final PDF

The markdown file you are reading (`docs/Project_Explained.md`) can be turned into a PDF with pandoc:

```
# From the repository root:
mkdir -p docs
pandoc docs/Project_Explained.md -o docs/Project_Explained.pdf
```

Pandoc will use the available PDF engine (wkhtmltopdf or any installed LaTeX backend). The resulting PDF will contain the full report, the updated disclaimer, and placeholders for the evaluation results. After you have run the two evaluation scripts, replace the placeholders in the markdown file with the actual numbers and re-run the `pandoc` command to obtain the final PDF.

End of report.